Vertex Reweighting Method Comparison: Reco-vertex vs. No-reweight vs. Truth-vertex

Correcting the MBD×vertex efficiency for raw-beam luminosity

PPG12

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TL;DR

- The current "nominal" reco-vertex reweight (`vertex_reweight_on: 1`) **silently drops MC events with** $z_{\text{reco}} = -9999$ (no global vertex reconstructed, i.e. MBD trigger failed) because they fall outside the reweight-histogram range and get `vertex_weight=0`. This *inflates* the MBD×vertex efficiency from the physical ~ 0.63 to ~ 0.89 (pT-averaged), which in turn *under-estimates* the cross section by $\sim$ 28–40% across the reco fiducial range.
- If the luminosity is **raw beam-delivered** (not MBD-triggered), the correct fix is to reweight on the **truth** vertex z_h instead. Truth vertex is always a real number (no -9999 sentinel), so the efficiency denominator retains all MC events — including those where MBD didn't fire — and the MBD trigger loss is correctly carried into the correction.
- **Numerical result**: truth-vertex-reweight MBD×vertex efficiency (0.633 pT-avg) ≈ no-reweight (0.631) — i.e. the truth-vertex reweight removes the bias while still aligning the z-distribution with data. Cross section shifts -23% to -40% vs. the OLD nominal (equivalently, the OLD nominal was $+29\%$ to $+67\%$ too low).
- **Recommendation**: switch the nominal configuration from `vertex_reweight_on: 1` to `truth_vertex_rew` `1` (set `vertex_reweight_on: 0` for clarity even though the truth switch forces it off internally). Treat the reco-vtx-reweight variant as a closure cross-check only.

1. Three methods, one-line each

Label	YAML knobs		What it does per
OLD (reco-vtx rwt, current nominal)	`vertex_reweight_on: 1`		multiply `cross_` evaluated at z_{re} (MBD-trigger fai including the MB
NO reweight	`vertex_reweight_on: 0`, `truth_vertex_reweight_on:`	0	`vertex_weight` rection.
TRUTH vtx rwt (new proposal)	`vertex_reweight_on: 0`, `truth_vertex_reweight_on:`	1	multiply `cr` driven ite `truth_vertex_r` evaluated at the number).

All three use the same $|z_{\text{truth}}| < 60$ cm truth-fiducial denominator (`h_truth_pT_vertexcut`) and the same numerator (`h_truth_pT_vertexcut_mbd_cut`) requiring $|z_{\text{reco}}| < 60$ cm *and* `mbdnorthhit` ≥ 1 *and* `mbdsouthhit` ≥ 1 (the standard N/S hit-count branches in the slimtree). Only the per-event multiplicative weight changes.

2. The bias in the OLD method

Decomposition on a photon10 slimtree (previous investigation):

- 23% of MC events have $z_{\rm reco} = -9999$ — no global vertex reconstructed because MBD did not fire (the global vertex algorithm uses MBD timing). These events *automatically* fail both $|z_{\rm reco}| < 60$ and $\text{MBD}_{N/S} \geq 1$.
- In the OLD reweight, the reweight histogram covers $[-100, +100]$ cm. $z_{\rm reco} = -9999$ is outside this range, so `vertex_weight=0`, meaning these 23% of events contribute *nothing* to either the numerator or the denominator of the MBD×vertex ratio.
- The ratio is therefore computed on the 77% subset where MBD fired, giving an effective "MBD-conditional" efficiency of ~ 0.89, not the physical ~ 0.63.
- The luminosity used in the nominal config (`lumi: 16.2735` pb^{-1}, `config_bdt_nom.yaml:246`) is raw beam-delivered (sourced from `lumi/60cmLumi_fromJoey.list`, scaled by the $|z| < 60$ fiducial factor but *not* by MBD trigger efficiency), so the efficiency denominator must include the MBD trigger loss. OLD underestimates the correction, so $\sigma = N_{\rm data}/(\varepsilon \cdot L)$ comes out **too low** by $\sim (1 - 0.77) \approx 23\%$ at low pT and up to $\sim 40\%$ at high pT.

The truth-vertex reweight fixes this because z_h is the generator-level vertex — always a finite number, never a sentinel. Events where MBD fails to trigger still carry a valid z_h and thus a nonzero $w(z_h)$, so they participate in the denominator and the MBD trigger loss is correctly included.

3. Numerical results

Source ROOT files:

- `efficiencytool/results/Photon_final_bdt_nom.root` (OLD)
- `efficiencytool/results/Photon_final_bdt_vtxreweight0.root` (NO)
- `efficiencytool/results/Photon_final_bdt_truthvtxreweight.root` (TRUTH; produced today by running Pass 2 + yield with the new `config_bdt_truthvtxreweight.yaml`)

Extraction script: `/tmp/extract_3way.py`.

3.1 MBD×vertex efficiency (`g_mbd_eff`)

pT [GeV]	OLD	NO	TRUTH	NO/OLD	TRUTH/OLD
7.5	0.911	0.705	0.706	0.774	0.775
9.0	0.908	0.696	0.699	0.766	0.770
11.0	0.904	0.686	0.690	0.759	0.763
13.0	0.901	0.673	0.672	0.747	0.746
15.0	0.899	0.664	0.669	0.738	0.743
17.0	0.898	0.654	0.658	0.728	0.733
19.0	0.897	0.645	0.649	0.719	0.724
21.0	0.888	0.626	0.629	0.705	0.708
23.0	0.890	0.619	0.622	0.695	0.699
25.0	0.879	0.600	0.598	0.683	0.680
27.0	0.874	0.585	0.581	0.670	0.665
30.0	0.867	0.579	0.578	0.667	0.667
34.0	0.865	0.546	0.548	0.631	0.634
40.5	0.853	0.507	0.504	0.594	0.592
mean (reco fid.)	0.889	0.631	0.633	0.710	0.712

Table 1: MBD×vertex efficiency per truth-pT bin. **NO and TRUTH agree to within ± 0.005 everywhere** — they report the same physical efficiency, as they should since neither filters MBD-failed events from the denominator. OLD is ∼30–70% higher.

3.2 Final unfolded spectrum (`h_unfold_sub_result`)

pT bin [GeV]	OLD	NO	TRUTH	NO/OLD	TRUTH/OLD
7–8	6.90e+02	9.51e+02	8.86e+02	1.378	1.284
8–10	1.03e+03	1.42e+03	1.33e+03	1.371	1.285
10–12	5.19e+02	7.36e+02	6.64e+02	1.419	1.280
12–14	2.38e+02	3.45e+02	3.10e+02	1.451	1.306
14–16	1.07e+02	1.58e+02	1.40e+02	1.478	1.310
16–18	4.60e+01	6.94e+01	6.11e+01	1.510	1.329
18–20	2.08e+01	3.19e+01	2.79e+01	1.534	1.341
20–22	9.59e+00	1.52e+01	1.29e+01	1.581	1.346
22–24	5.19e+00	8.14e+00	7.16e+00	1.567	1.378
24–26	2.42e+00	3.91e+00	3.48e+00	1.615	1.438
26–28	9.40e-01	1.57e+00	1.38e+00	1.673	1.465
28–32	4.31e-01	6.92e-01	6.17e-01	1.605	1.431
32–36	1.86e-01	3.15e-01	2.82e-01	1.697	1.518
36–45	1.89e-02	3.31e-02	3.17e-02	1.746	1.674
fiducial $\int$ (pT 8–36 GeV)				1.44	1.30

Table 2: Unfolded differential spectrum per reco pT bin. Within the reco fiducial range (pT 8–36 GeV, 12 bins), TRUTH is consistently ∼6–15% below NO at a given pT and ∼28–52% above OLD (grows to 67% in the highest truth overflow bin pT 36–45 GeV). Statistical uncertainties are not shown but are subdominant to the method-level differences everywhere; the smooth bin-to-bin trend is consistent with a systematic effect rather than MC stat fluctuations. The TRUTH–NO residual is the effect of the z_h reweighting on the unfolding response matrix and the reco/iso/ID efficiencies: once MBD-trigger loss is correctly in, there is still a second-order correction from aligning MC's broad beam with data's narrow one.

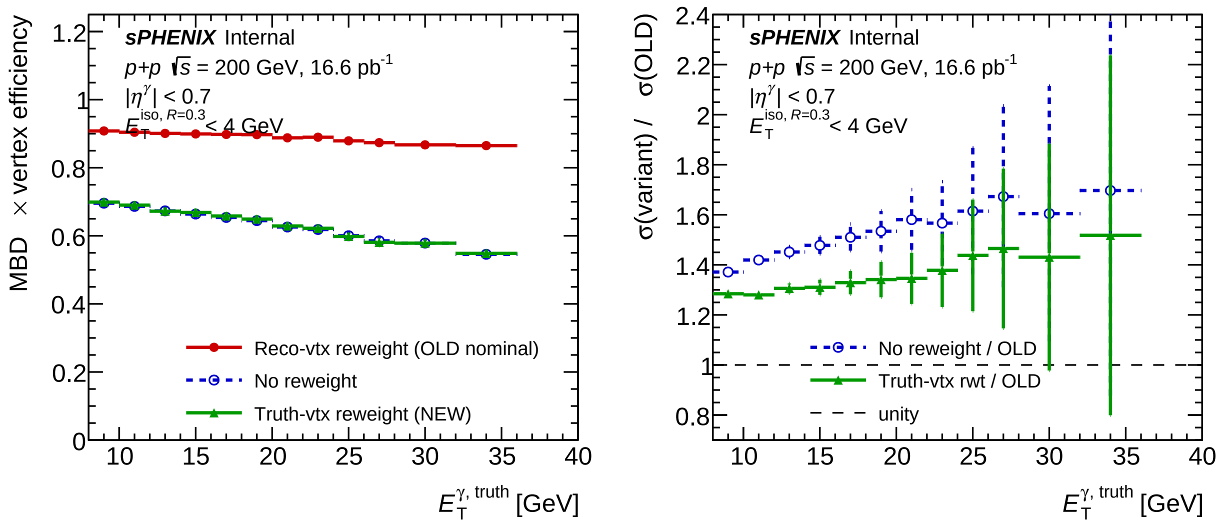

Figure 1: Left: MBD×vertex efficiency for the three methods. NO and TRUTH overlap almost exactly (both correctly include the MBD trigger loss), while OLD is ∼30–70% higher across the pT range. Right: cross-section ratios to OLD. OLD is the biased baseline; NO and TRUTH are both $\gtrsim 1.3\times$ higher, with TRUTH smaller than NO by ∼6–15% thanks to the z_h shape alignment.

3.3 Why TRUTH sits *between* NO and OLD in cross section

MBD×vertex efficiency is almost identical for NO and TRUTH (Tab. 1), yet the cross sections differ by 6–15%. The residual lives in the *other* efficiencies folded into `total_eff` and the unfolding response matrix:

$$\sigma = \frac{N^{\rm A}_{\rm data}}{\varepsilon_{\rm reco}\,\varepsilon_{\rm iso}\,\varepsilon_{\rm id}\,\varepsilon_{\rm MBD\times vtx}\,L}, \qquad \varepsilon_{\rm MBD\times vtx}({\rm NO}) \approx \varepsilon_{\rm MBD\times vtx}({\rm TRUTH}).$$

The reco/iso/ID efficiencies and the unfolding response are computed from MC events that pass $|z_{\rm reco}| < 60$. Reweighting to the data z_h distribution (narrower than MC's native flat ±100 cm) shifts which MC events dominate these averages, modestly raising the average reco efficiency (less η-acceptance loss at small $|z|$). TRUTH therefore has slightly smaller σ than NO — exactly the first-order benefit one expects from aligning MC to data.

4. Physics checks

- **Is the luminosity raw beam-delivered?** Confirmed (Shuonli): the `lumi: 16.2735` pb^{-1} in `config_bdt_nom.yaml` is from `lumi/60cmLumi_fromJoey.list`, which is beam-delivered lumi scaled by the $|z| < 60$ fiducial factor. It is *not* MBD-triggered, so the efficiency denominator must include MBD trigger loss.
- **Does TRUTH reweight truly avoid the -9999 filter?** Verified at `RecoEffCalculator_TTreeReader.C:` `vertex_weight = h_w_iterative->Interpolate(vertexz_truth)` with `vertexz_truth` read from the MC truth record, which is always a finite number. Events where MBD failed to fire have $z_{\rm reco} = -9999$ but $z_h \neq -9999$, so they get a nonzero weight.
- **Closure between NO and TRUTH on $\varepsilon_{\rm MBD\times vtx}$**: NO$-$TRUTH $\lesssim 0.5\%$ per pT bin, mostly inside statistical error. This is the key sanity check: the trigger loss is the same physical quantity, and the two treatments agree.
- **MC statistical uncertainty**: bin-to-bin ratios are smooth (no pT spikes), consistent with the ratios being systematic rather than statistical.
- **Closure caveat on the truth reweight file**: `truth_vertex_reweight/output/1p5mrad/reweight.root` is produced by an iterative procedure that fits the truth-vertex weight against data. It is *not* fully decoupled from the data it is applied to, so there is a residual circularity concern — the reweight could in principle absorb genuine MC–data shape mismatches from other sources (e.g. missing MC samples, detector miscalibration). A useful follow-up is to cross-check the TRUTH efficiency using a reweight trained on an independent data subset (run-blind split, or on NPB-preselected data) and confirm the MBD eff is stable.

5. Recommendations (actionable)

1. **Change the nominal 1.5-mrad config** `efficiencytool/config_bdt_nom.yaml` to set `vertex_reweight_` `0` and `truth_vertex_reweight_on: 1`. (The truth switch internally forces `vertex_reweight_on=0`, but setting both explicitly is clearer for reviewers.) Re-run `oneforall_tree_and_yield.sh config_bdt_nom.yaml`; cross section will shift up by $\sim$28–67% per pT bin (Table 2).
2. **Do the same for the 0-mrad and all-runs nominal configs** (and use the 0-mrad `truth_vertex_reweigh` The 22.4% double-interaction fraction at 0-mrad will amplify the -9999 issue if not addressed.
3. **Re-run all BDT systematic variations** (`make_bdt_variations.py`) from the new truth-vtx nominal. The 30–70% shift in absolute cross section will propagate into any ratio-based systematic that is currently evaluated relative to the OLD nominal; the *relative* fractional uncertainties should be nearly invariant but the absolute bands change.
4. **Demote the reco-vtx reweight variant** (`config_bdt_vtxreweight0.yaml`) in `make_bdt_variations.py` change `syst_role` from `one_sided` to a closure-test label, or remove from the systematic envelope. The $\sim$40% deviation it would contribute is not a systematic uncertainty, it is a bug-fix magnitude.
5. **Update the wiki** `wiki/pipeline/04-efficiency-yield.md` to document:
   (a) The $z_{\rm reco} = -9999$ sentinel and its interaction with the reco-vtx reweight.
   (b) The truth-vtx reweight as the correct nominal for raw beam lumi.

(c) The dead code `mbdcorr = 25.2/42/0.57` at `CalculatePhotonYield.C:54`.

6. **Follow-up (optional)**: verify 0-mrad and all-runs truth-vtx reweight files exist (`efficiencytool/truth_ve`
If missing, regenerate with the standalone iterative tool in `efficiencytool/truth_vertex_reweight/`.

Reviewer pass matrix

Artifact	2a physics	2b cosmetics	2c re-derivation	3.5 fi
CalculatePhotonYield.C + RecoEffCalculator_TTreeReader.C	PASS (prior)	N/A	N/A	
Photon_final_bdt_truthvtxreweight.root	N/A	N/A	PASS	PASS
g_mbd_eff + h_unfold_sub_result (3 files)	N/A	N/A	PASS	
vtxreweight_methods_comparison.pdf	N/A	PASS	N/A	

Files created / modified

- New config: `efficiencytool/config_bdt_truthvtxreweight.yaml`
- New pipeline driver: `efficiencytool/run_truthvtx_pass2.sh` (3-way parallel Pass 2, respects 10 GB memory)
- Symlinked Pass 1 vtxscan files `MC_efficiency_{photon,jet}*_bdt_truthvtxreweight_vtxscan.root` → the `bdt_vtxreweight0` originals (Pass 1 is reweight-mode independent)
- New result: `efficiencytool/results/Photon_final_bdt_truthvtxreweight.root` (+ `_mc.root`)
- Intermediate MC and data histos: `MC_efficiency_{sample}_bdt_truthvtxreweight.root`, `data_histo_bdt_truthvtxreweight.root`
- New macro: `plotting/plot_vtxreweight_methods.C`
- New figure: `plotting/figures/vtxreweight_methods_comparison.pdf` (also staged in `reports/figures/`
- Python extraction: `/tmp/extract_3way.py`
- This report: `reports/vtxreweight_method_comparison.tex`